Unwelcome Guests and Their Even Smaller Guests

Do Marine Invasions Alter Native Parasite Communities in Puget Sound?

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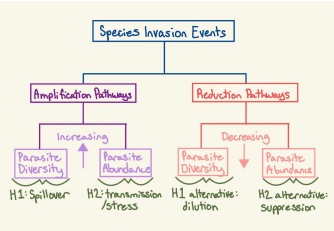
Introduction

- Parasites play key roles in food webs and host population dynamics but are rarely included in long-term ecological monitoring.
- Marine invasions can restructure host communities and disease transmission, potentially altering parasite diversity and abundance.
- Puget Sound offers rare opportunity to test invasion-parasite relationships using long-term museum data.



Hypotheses

H1: Spillover Hypothesis (Diversity Increase)
Increased invasion pressure will be associated with higher parasite diversity in native fishes.
H1 Alternative: Dilution Hypothesis (Diversity Decrease)
Invasion pressure will reduce parasite diversity by disrupting native transmission pathways.
H2: Transmission/Stress Hypothesis (Abundance Increase)
Increased invasion pressure will be associated with higher parasite abundance in native fishes.
H2 Alternative: Suppression Hypothesis (Abundance Decrease)
Parasite abundance will decline during years of elevated invasion activity.

Methods

- Compiled multi-decade parasite records from native Puget Sound fishes using museum collections.
- Defined invasion pressure as the number of non-native fish species first detected per year.
- Calculated annual parasite abundance (mean per host) and parasite diversity.
- Tested relationships using correlations, linear models, negative binomial GLMs, and 3-year lag models.
- Accounted for sampling effort using annual specimen counts.

Results

- No sig. relationship between invasion pressure and parasite abundance.
- No sig. relationship between invasion pressure and parasite diversity.
- No delayed (3-year lag) effects of invasions on parasite abundance.
- No delayed (3-year lag) effects of invasions on parasite diversity.
- Patterns were consistent across all statistical approaches.

Discussion

- Results do not support spillover or transmission/stress hypotheses.
- Results also do not support dilution or suppression hypotheses.
- Native parasite communities may be resilient to invasion-driven disturbance at the ecosystem scale.
- Invasion impacts may be species-specific or masked when analyzed at broad spatial and temporal scales.

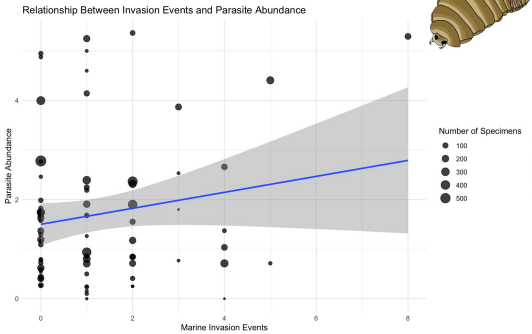


Figure 1: Relationship between the number of newly established non-native marine species recorded per year and mean parasite abundance in native fishes. No significant relationship detected.

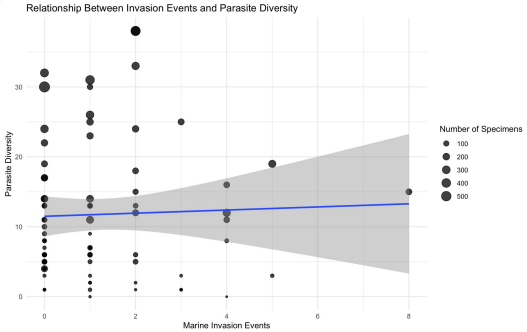


Figure 2: Relationship between the number of newly established non-native marine species recorded per year and parasite diversity in native fishes. No significant relationship detected.

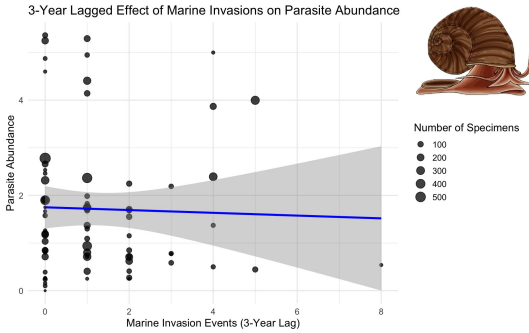


Figure 3: Relationship between the number of newly established non-native marine species recorded three years earlier and mean parasite abundance in native fishes. No significant relationship detected.

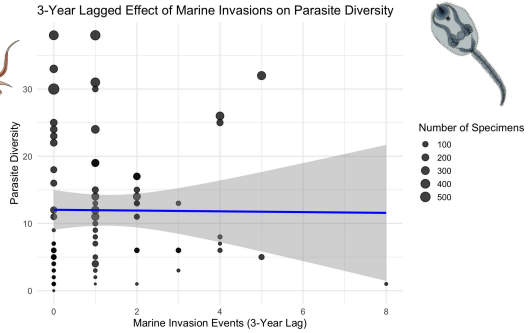


Figure 4: Relationship between the number of newly established non-native marine species recorded three years earlier and parasite diversity in native fishes.. No significant relationship detected.

Conclusion:

- Marine invasion activity in Puget Sound does not predict long-term parasite abundance or diversity in native fishes.
- Variable museum sampling and coarse invasion metrics may mask host-specific responses
- Future work should examine species-level and trophic transmission pathways.